

Technology Stack

Date	04 July 2026
Team ID	xxxxxx
Project Name	Opticrop-Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript, Bootstrap	<i>Provides a responsive and user-friendly interface for farmers to interact with the system on both mobile and desktop devices.</i>

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Flask is lightweight and efficient for developing machine learning web applications and handling prediction requests.</i>
3	Database / Data Storage	<i>MySQL</i>	<i>Stores crop, soil, weather, and user data securely with strong relational data management capabilities.</i>
4	Cloud / Hosting / Deployment	<i>AWS / Render / Heroku</i>	<i>Enables scalable and reliable deployment of the OptiCrop application with easy accessibility from anywhere.</i>
5	Version Control & CI/CD	<i>GitHub</i>	<i>Supports source code management, collaboration, version tracking, and continuous development.</i>

6	Third-Party APIs / Other Tools	<i>OpenWeatherMap API, Scikit-learn, Pandas, NumPy</i>	<i>OpenWeatherMap provides real-time weather data; Scikit-learn is used for crop prediction models; Pandas and NumPy assist in data processing and analysis.</i>